PHYS 6031
Spring 2024
Homework 6
Due: Tuesday 8 March 2024 at 4:00 pm in my Clippinger mailbox

1. A metal cylinder of radius a and length L has both ends held at a zero potential, and the sides are at constant potential V_0.

(a) Find the potential $\Phi(\rho, \phi, z)$ everywhere in the interior of the cylinder. Assume the cylinder is aligned with the z axis and that and that the ends are located at $z = 0$ and $z = L$. I suggest choosing a functional form that vanishes explicitly for $z = L$ and $z = L$.

(b) Now consider the same question, but with some shifts applied:

$$\Phi'(\rho, \phi, z') = \Phi(\rho, \phi, z + L/2) - V_0,$$

i.e., such that the center of the cylinder is now at the origin, the potential vanishes on the sides of the cylinder, and the potential is equal to $-V_0$ on the ends. Find Φ' using functions that vanish explicitly on the sides of the cylinder.

(c) Which solution of the problem is "better" ? Why?

2. The current distribution of an infinite, longitudinally-uniform, and axially-symmetric wire can be described in cylindrical coordinates by

$$\vec{\boldsymbol{J}} = J(\rho)\hat{\boldsymbol{z}}.$$

(a) Show that $\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{J}} = 0$.

(b) Considerations of longitudinal and axial symmetry require that the magnetic field can only depend upon ρ, i.e. that

$$\vec{\boldsymbol{B}}(\rho) = B_\rho(\rho)\hat{\boldsymbol{\rho}} + B_\phi(\rho)\hat{\boldsymbol{\phi}} + B_z(\rho)\hat{\boldsymbol{z}}.$$

Use Ampère's Law to determine $B_\phi(\rho)$ in terms of $I(\rho) \equiv 2\pi \int_0^\rho J(\rho')\, \rho' \mathrm{d}\rho'$.

(c) Use the Biot and Savart Law to show that $B_\rho(\rho) = B_z(\rho) = 0$. Side note: it is pretty easy to show that $B_\rho(\rho)$ must be zero using Gauss's Law for $\vec{\boldsymbol{B}}$ with a cylindrical volume. I am not aware of an "easy" way to see $B_z(\rho) = 0$ – not that it is very difficult using the Biot and Savart Law...

(d) Use the integral form for the vector potential [Jackson, Eq. (5.32)] to determine $\vec{\boldsymbol{A}}$ for this current distribution. Hint: in order to deal with divergent integrals, you may want to limit the current distribution to $-L \leq z \leq L$, determine $\vec{\boldsymbol{A}}$, adjust $\vec{\boldsymbol{A}}$ by a constant (as one is allowed to do), and then take $L \to \infty$.

(e) Verify that the $\vec{\boldsymbol{A}}$ in part (d) reproduces the $\vec{\boldsymbol{B}}$ found in parts (b) and (c) when the curl is taken.

3. The combination of two circular, coaxial coils of radius a separated by a distance also equal to a is known as a pair of "Helmholtz Coils." The current distribution can be described in cylindrical coordinates as:

$$\vec{\boldsymbol{J}} = I\delta(\rho - a)[\delta(z - a/2) + \delta(z + a/2)]\hat{\boldsymbol{\phi}}.$$

(a) Starting from Jackson, Eq. (5.32) find $\vec{\boldsymbol{A}}$ for all space. I suggest working in *spherical* coordinates, and using an approach similar to section 5.5 of Jackson.

(b) Find the leading non-constant behavior of the z component of $\vec{\boldsymbol{B}}$ near the origin, along the z axis. I suggest using an elementary approach here (i.e., Biot-Savart Law), rather than the formula derived above.

Note: Helmholtz Coils are commonly used in experimental physics to generate a known and very uniform magnetic field.